Cryptographic Techniques

Classical Encryption Techniques

- The Classical Encryption Techniques are
  - Substitution Technique
  - Transposition Technique
- In substitution technique, letters are replaced by other letters or symbols.

Example:

a	b	c	d	e	f	g	h	i	j	k	l	m
n	o	p	q	r	s	t	u	v	w	x	y	z

a → M

b → X

x → Z

g → A

Plaintext : bag

Ciphertext : XMA

Classical Encryption Techniques

- In transposition technique, applying some sort of permutation on the plaintext letters.
- Example:
  Plaintext: ACER
  Ciphertext: CERA, ERAC, RACE, CARE, etc.,

Classical Encryption Techniques

Substitution	Transposition
Caesar Cipher	Rail Fence
Monoalphabetic Cipher	Row Column Transposition
Playfair Cipher	
Hill Cipher	
Polyalphabetic Cipher	
One-Time Pad	

Figure 1 : Different Classical Encryption Techniques

Caesar Cipher

Caesar Cipher

- Caesar Cipher: Letters are replaced by other letters.
- Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

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- Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

Example:

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher

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- Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

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A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Algorithm:

For each plaintext letter 'p', substitute the ciphertext letter 'C':

$C = E(p, k) \bmod 26 = (p + k) \bmod 26$

$p = D(C, k) \bmod 26 = (C - k) \bmod 26$

Caesar Cipher

- Advantages:
  - Simple
  - Easy to implement
- Disadvantages:
  - The encryption and decryption algorithms are known
  - There are only 25 keys to try (Vulnerable to brute force attack)

Caesar Cipher

Ciphertext: SQDYMZK

Shifts	Back	Result	Shifts	Back	Result
0	[26]	SQDYMZK	13	[13]	FDQLZMX
1	[25]	TREZNAL	14	[12]	GERMANY
2	[24]	USFAOBM	15	[11]	HFSNBOZ
3	[23]	VTGBPCN	16	[10]	IGTOCPA
4	[22]	WUHCQDO	17	[9]	JHUPDQB
5	[21]	XVIDREP	18	[8]	KIVQERC
6	[20]	YWJESFQ	19	[7]	LJWRFSD
7	[19]	ZXKFTGR	20	[6]	MKXSGTE
8	[18]	AYLGUHS	21	[5]	NLYTHUF
9	[17]	BZMHVIT	22	[4]	OMZUIVG
10	[16]	CANIWJU	23	[3]	PNAVJWH
11	[15]	DBOJXKV	24	[2]	QOBWKXI
12	[14]	ECPKYLW	25	[1]	RPCXLYJ
13	[13]	FDQLZMX			

Figure 2 : Brute force attack

Playfair Cipher

Playfair Cipher

- Playfair Cipher: Invented by Charles Wheatstone.
- Manual symmetric encryption technique.
- Multiple letter encryption cipher.

Playfair Cipher

- 5×5 matrix constructed using a keyword.
  Example: Monarchy

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher

- Rules for encryption using Playfair cipher:
  - Create digrams
  - Repeating letters - Filler letter
  - Same column | $\downarrow$ | Wrap around
  - Same row | $\rightarrow$ | Wrap around
  - Rectangle | $\leftrightarrow$ | Swap

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  - Create digrams
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- Example1:

Plaintext: attack

Digrams: at ta ck

Playfair Cipher

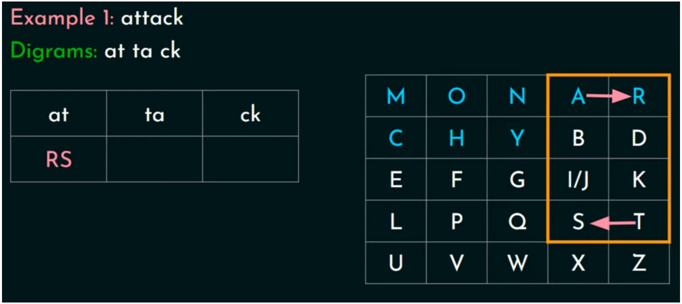
- Rules for encryption using Playfair cipher:
  - Create digrams
  - Repeating letters - Filler letter
  - Same column | $\downarrow$ | Wrap around
  - Same row | $\rightarrow$ | Wrap around
  - Rectangle | $\leftrightarrow$ | Swap
- Example1:

Plaintext: attack
Digrams: at ta ck

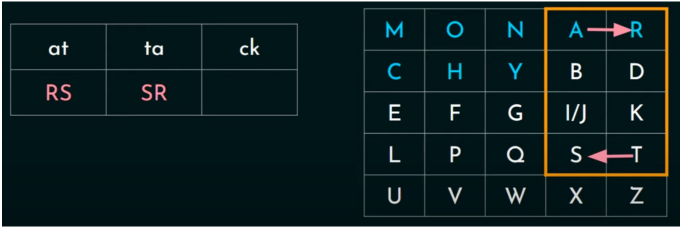
- Example2:

Plaintext: balloon
Digrams: ba ll oo n
Digrams: ba lx lo on

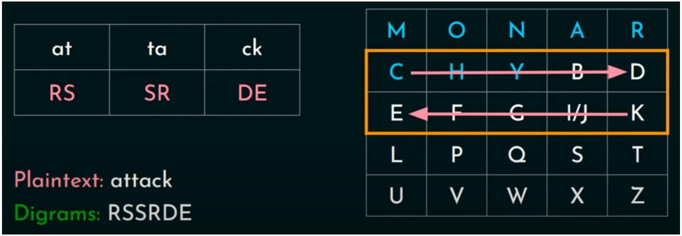
Playfair Cipher



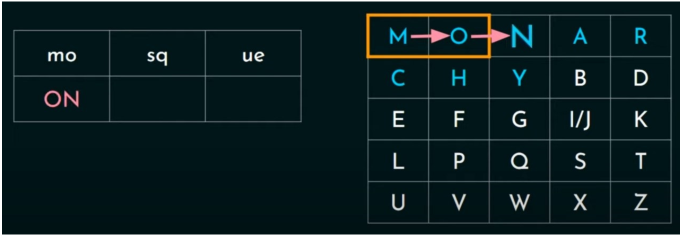
Playfair Cipher



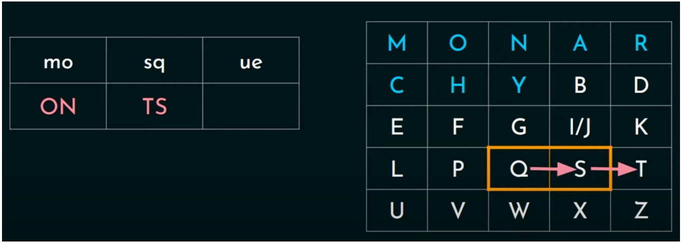
Playfair Cipher



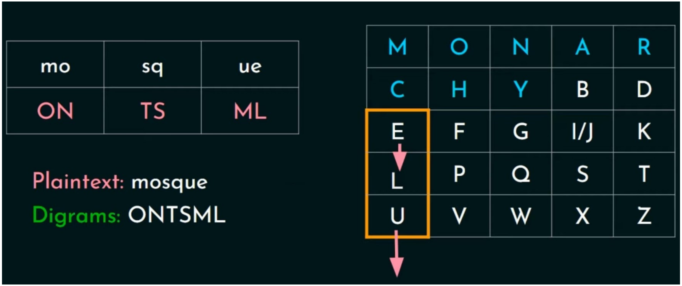
Playfair Cipher



Playfair Cipher



Playfair Cipher



Playfair Cipher

- Exercise1: Encrypt the message "STOP THE WAR" using Playfair cipher. Assume the keyword as "KEYWORD"

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  Ciphertext: MZYSVFYOBD

Playfair Cipher

- Exercise2: Decrypt the Ciphertext "QLPQLIIZBA" using Playfair cipher. Assume the keyword as "HITLER"

Playfair Cipher

- Exercise2: Decrypt the Ciphertext "QLPQLIIZBA" using Playfair cipher. Assume the keyword as "HITLER"
  Ciphertext: STOP THE WAR

Hill Cipher

Hill Cipher

- Hill Cipher: Developed by Lester Hill in 1929
- Multi-letter cipher
- Encrypts a group of letters: digraph, trigraph or polygraph
- Concepts
  - Matrix arithmetic modulo 26
  - Square matrix
  - Determinant
  - Multiplicative inverse

Hill Cipher

- The Hill Algorithm:

$$C = E(K,P) = P \times K \bmod 26$$

$$P = D(K,C) = C\,K^{-1} \bmod 26 = P \times K \times K^{-1} \bmod 26$$

Hill Cipher

- The Hill Algorithm:

$$C = E(K,P) = P \times K \bmod 26$$

$$P = D(K,C) = C\,K^{-1} \bmod 26 = P \times K \times K^{-1} \bmod 26$$

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26 \quad \text{Encryption}$$

$$C_1 = (P_1K_{11} + P_2K_{21} + P_3K_{31}) \bmod 26$$
$$C_2 = (P_1K_{12} + P_2K_{22} + P_3K_{32}) \bmod 26$$
$$C_3 = (P_1K_{13} + P_2K_{23} + P_3K_{33}) \bmod 26$$

Hill Cipher

- Exercise1: Encrypt the message "pay more money" using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Hill Cipher

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$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Solution:

p	a	y	m	o	r	e	m	o	n	e	y
15	0	24	12	14	17	4	12	14	13	4	24

Key = 3 x 3 matrix.

PT = pay mor emo ney

Hill Cipher

Encrypting: pay

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$\begin{aligned}(C_1\ C_2\ C_3) &= (15\ 0\ 24)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26 \\ &= (15\text{x}17+0\text{x}21+24\text{x}2 \quad 15\text{x}17+0\text{x}18+24\text{x}2 \quad 15\text{x}5+0\text{x}21+24\text{x}19) \bmod 26 \\ &= (303\ 303\ 531) \bmod 26 \\ &= (17\ 17\ 11) \\ &= (R\ R\ L)\end{aligned}$$

Hill Cipher

Encrypting: mor

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$\begin{aligned}(C_1\ C_2\ C_3) &= (12\ 14\ 17)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26 \\ &= (12\text{x}17+14\text{x}21+17\text{x}2 \quad 12\text{x}17+14\text{x}18+17\text{x}2 \quad 12\text{x}5+14\text{x}21+17\text{x}19) \bmod 26 \\ &= (532\ 490\ 677) \bmod 26 \\ &= (12\ 22\ 1) \\ &= (\text{M W B})\end{aligned}$$

Hill Cipher

Encrypting: emo

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (4\ 12\ 14)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$= (4\text{x}17+12\text{x}21+14\text{x}2 \quad 4\text{x}17+12\text{x}18+14\text{x}2 \quad 4\text{x}5+12\text{x}21+14\text{x}19) \bmod 26$$

$$= (348\ 312\ 538) \bmod 26$$

$$= (10\ 0\ 18)$$

$$= (K\ A\ S)$$

Hill Cipher

Encrypting: ney

$$(C_1\ C_2\ C_3) = (P_1\ P_2\ P_3)\begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

$$\begin{aligned}(C_1\ C_2\ C_3) &= (13\ 4\ 24)\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26 \\ &= (13x17+4x21+24x2 \quad 13x17+4x18+24x2 \quad 13x5+4x21+24x19) \bmod 26 \\ &= (348\ 312\ 538) \bmod 26 \\ &= (15\ 3\ 7) \\ &= (P\ D\ H)\end{aligned}$$

Hill Cipher

PT	p	a	y	m	o	r	e	m	o	n	e	y
CT	R	R	L	M	W	B	K	A	S	P	D	H

Hill Cipher

- Decryption requires K^{-1}, the inverse matrix of K.

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

Hill Cipher

- Decryption requires K^{-1}, the inverse matrix of K.

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

To find the determinant of K: $\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$

$$\text{Det} \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{ mod } 26$$

$$
\begin{aligned}
&= 17(18\text{x}19 - 2\text{x}21) - 17(19\text{x}21 - 2\text{x}21) + 5(2\text{x}21 - 2\text{x}18) \text{ mod } 26 \\
&= 17(342 - 42) -17(399 - 42)+5(42 - 36) \text{ mod } 26 \\
&= 17(300) - 17(357)+ 5(6) \text{ mod } 26 \\
&= 5100 - 6069 + 30 \text{ mod } 26 \\
&= -939 \text{ mod } 26 \\
&= -3 \text{ mod } 26 \\
&= 23
\end{aligned}
$$

Hill Cipher

To find Adjoint K

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

$$\text{Adj } K = \begin{vmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{vmatrix}$$

$$\text{Adj } K = \begin{matrix} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{matrix}$$

Hill Cipher

$$\text{Adj } K = \begin{array}{ccccc} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{array}$$

$$\text{Adj } K = \begin{array}{ccccc} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{array}$$

$$\text{Adj } K = \begin{array}{ccccc} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \end{array}$$

$$\begin{array}{ccccc} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \end{array}$$

$$
\text{Adj } K = \begin{array}{ccccc}
17 & 17 & 5 & 17 & 17 \\
21 & 18 & 21 & 21 & 18 \\
2 & 2 & 19 & 2 & 2 \\
 & & & & \\
17 & 17 & 5 & 17 & 17 \\
21 & 18 & 21 & 21 & 18
\end{array}
$$

$$\text{Adj } K = \begin{array}{ccccc} 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \\ 2 & 2 & 19 & 2 & 2 \\ 17 & 17 & 5 & 17 & 17 \\ 21 & 18 & 21 & 21 & 18 \end{array}$$

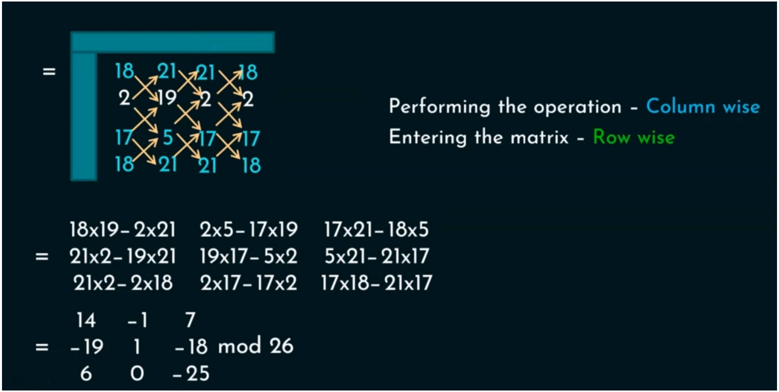
$$= \begin{array}{cccc} 18 & 21 & 21 & 18 \\ 2 & 19 & 2 & 2 \\ 17 & 5 & 17 & 17 \\ 18 & 21 & 21 & 18 \end{array}$$

Performing the operation - Column wise

Entering the matrix - Row wise

$$= \begin{array}{ccc} 18\text{x}19 - 2\text{x}21 & 2\text{x}5 - 17\text{x}19 & 17\text{x}21 - 18\text{x}5 \\ 21\text{x}2 - 19\text{x}21 & 19\text{x}17 - 5\text{x}2 & 5\text{x}21 - 21\text{x}17 \\ 21\text{x}2 - 2\text{x}18 & 2\text{x}17 - 17\text{x}2 & 17\text{x}18 - 21\text{x}17 \end{array}$$

$$= \begin{array}{ccc} 300 & -313 & 267 \\ -357 & 313 & -252 \\ 6 & 0 & -51 \end{array} \bmod 26$$


=
18 21 21 18
2 19 2 2
17 5 17 17
18 21 21 18
Performing the operation - Column wise
Entering the matrix - Row wise
18x19 - 2x21   2x5 - 17x19   17x21 - 18x5
= 21x2 - 19x21   19x17 - 5x2   5x21 - 21x17
21x2 - 2x18   2x17 - 17x2   17x18 - 21x17
14   -1   7
= -19   1   -18 mod 26
6   0   -25

$$= \begin{array}{cccc} 18 & 21 & 21 & 18 \\ 2 & 19 & 2 & 2 \\ 17 & 5 & 17 & 17 \\ 18 & 21 & 21 & 18 \end{array}$$

Performing the operation - Column wise

Entering the matrix - Row wise

$$= \begin{array}{ccc} 18x19-2x21 & 2x5-17x19 & 17x21-18x5 \\ 21x2-19x21 & 19x17-5x2 & 5x21-21x17 \\ 21x2-2x18 & 2x17-17x2 & 17x18-21x17 \end{array}$$

$$= \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

Hill Cipher

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

$$K^{-1} = \frac{1}{23} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

$$K^{-1} = 23^{-1} \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

Hill Cipher

$23^{-1} \times 23 = 1 \bmod 26$	$9 \times 23 = 25 \bmod 26$
$1 \times 23 = 23 \bmod 26$	$10 \times 23 = 22 \bmod 26$
$2 \times 23 = 20 \bmod 26$	$11 \times 23 = 19 \bmod 26$
$3 \times 23 = 17 \bmod 26$	$12 \times 23 = 16 \bmod 26$
$4 \times 23 = 14 \bmod 26$	$13 \times 23 = 13 \bmod 26$
$5 \times 23 = 11 \bmod 26$	$14 \times 23 = 10 \bmod 26$
$6 \times 23 = 8 \bmod 26$	$15 \times 23 = 7 \bmod 26$
$7 \times 23 = 5 \bmod 26$	$16 \times 23 = 4 \bmod 26$
$8 \times 23 = 2 \bmod 26$	$17 \times 23 = 1 \bmod 26$

Hill Cipher

$$K^{-1} = 17 \times \begin{pmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{pmatrix} \bmod 26$$

$$K^{-1} = \begin{pmatrix} 238 & 425 & 119 \\ 119 & 17 & 136 \\ 102 & 0 & 17 \end{pmatrix} \bmod 26$$

$$K^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

Hill Cipher

$$K \times K^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$K = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \quad K^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix}$$

This is demonstrated as

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} = \begin{pmatrix} 443 & 442 & 442 \\ 858 & 495 & 780 \\ 494 & 52 & 365 \end{pmatrix} \bmod 26$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Hill Cipher

Decrypting: RRL

$$(P_1\ P_2\ P_3) = (R\ R\ L)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

Decryption

$$(C_1\ C_2\ C_3) = (17\ 17\ 14)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$= (17\times4+17\times15+11\times24 \quad 17\times9+17\times17+11\times0 \quad 17\times15+17\times6+11\times17) \bmod 26$$

$$= (587\ 442\ 544) \bmod 26$$

$$= (15\ 0\ 24)$$

$$= (P\ A\ Y)$$

Hill Cipher

Decrypting: MWB

$$(P_1\ P_2\ P_3) = (M\ W\ B)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$\begin{aligned}(C_1\ C_2\ C_3) &= (12\ 22\ 1)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26 \\ &= (12\text{x}4+22\text{x}15+1\text{x}24 \quad 12\text{x}9+22\text{x}17+1\text{x}0 \quad 12\text{x}15+22\text{x}6+1\text{x}17) \bmod 26 \\ &= (402\ 482\ 329) \bmod 26 \\ &= (12\ 14\ 17) \\ &= (M\ O\ R)\end{aligned}$$

Hill Cipher

Decrypting: KAS

$$(P_1\ P_2\ P_3) = (K\ A\ S)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$\begin{aligned}(C_1\ C_2\ C_3) &= (10\ 0\ 18)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26 \\ &= (10\text{x}4+0\text{x}15+18\text{x}24 \quad 10\text{x}9+0\text{x}17+18\text{x}0 \quad 10\text{x}15+0\text{x}6+18\text{x}17) \bmod 26 \\ &= (472\ 90\ 456) \bmod 26 \\ &= (4\ 12\ 14) \\ &= (E\ M\ O)\end{aligned}$$

Hill Cipher

Decrypting: PDH

$$(P_1\ P_2\ P_3) = (P\ D\ H)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$(C_1\ C_2\ C_3) = (15\ 3\ 7)\begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \bmod 26$$

$$= (15\text{x}4+3\text{x}15+7\text{x}24 \quad 15\text{x}9+3\text{x}17+7\text{x}0 \quad 15\text{x}15+3\text{x}6+7\text{x}17) \bmod 26$$

$$= (273\ 186\ 362) \bmod 26$$

$$= (13\ 4\ 24)$$

$$= (N\ E\ Y)$$

Hill Cipher

CT	R	R	L	M	W	B	K	A	S	P	D	H
PT	p	a	y	m	o	r	e	m	o	n	e	y

Hill Cipher

- Exercise1: Encrypt the plaintext "attack" using Hill Cipher for the given key

$$\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix}$$

Hill Cipher

- Exercise1: Encrypt the plaintext "attack" using Hill Cipher for the given key

$$\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix}$$

Ciphertext: FKMFIO

Hill Cipher

- Exercise2: Decrypt the ciphertext "FKMFIO" using Hill Cipher for the given key

$$\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix}$$

Hill Cipher

- Exercise2: Decrypt the ciphertext "FKMFIO" using Hill Cipher for the given key

$$\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix}$$

Plaintext: attack

Polyalphabetic Cipher

Polyalphabetic Cipher

- Polyalphabetic Cipher: To improve on the simple monoalphabetic technique
- Common Features:
  - A set of related monoalphabetic substitution rules is used
  - A key determines which particular rule is chosen for a given transformation.

Polyalphabetic Cipher

- Vignere Cipher:
  It consists of the 26 Caesar ciphers with shift 0 through 25.

Encryption process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Decryption process:

$$P_i = (C_i - K_{i \bmod m}) \bmod 26$$

Vignere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17
CT	25	8	2	21	19	22	16	13	6	17	25	6	21

Key	15	19	8	21	4	3	4	2	4	15	19	8	21	4
PT	4	3	18	0	21	4	24	14	20	17	18	4	11	5
CT	19	22	0	21	25	7	2	16	24	6	11	12	6	9

Thank you

&

Queries